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CICADA 3301 · LIBER PRIMUS 0-2



# SOLUTION 0-2

Volume 1

27x27 Rune Matrix · Three-Rune Nodes · Euler Totients · Mobius Phase

Current proposed plaintext

*AS I GO, THE WEATHER TURNS COLD.*

A cleaned reader edition of the original Volume 1 research. The order and wording stay close to the source; the presentation has been reorganized for clarity.

# A possible key before the matrix

Volume 1 begins with a small word construction before the 27x27 route. Using the same o-based Gematria Primus values:

ROAD = R(4) + O(3) + A(24) + D(23) = 54 -> 54 mod 29 = 25 -> AE PATH = P(13) + A(24) + TH(2) = 39 -> 39 mod 29 = 10 -> I WAY = W(7) + A(24) + Y(26) = 57 -> 57 mod 29 = 28 -> EA

This gives AE-I-EA. The same key later appears inside the matrix.

## Recursive Transition Networks

The original Volume 1 also opens with pages 139–140 of Gödel, Escher, Bach on Recursive Transition Networks. In the source this is conceptual context: a route can move through states, call another structure, and return. It is not required for the arithmetic itself, but it influenced the way the map-like structure was interpreted.

139–140 of GEB

### Recursive Transition Networks

The syntactical structure of sentences affords a good place to present a of describing recursive structures and processes: the *Recursive Transition Network (RTN)*. An RTN is a diagram showing various paths which can be followed to accomplish a particular task. Each path consists of a number of *nodes*, or little boxes with words in them, joined by *arcs*, or lines with arrows. The overall name for the RTN is written separately at the left, and the and last nodes have the words *begin* and *end* in them. All the other nodes contain either very short explicit directions to perform, or else name other RTN's. Each time you hit a node, you are to carry out the direct inside it, or to jump to the RTN named inside it, and carry it out.

Let's take a sample RTN, called **ORNATE NOUN**, which tells how to construct a certain type of English noun phrase. (See Fig. 27a.) If traverse **ORNATE NOUN** purely horizontally, we *begin'*, then we create **ARTICLE**, an **ADJECTIVE**, and a **NOUN**, then we *end*. For instance, "the shampoo" or "a thankless brunch". But the arcs show other possibilities such as skipping the article, or repeating the adjective. Thus we co construct "milk", or "big red blue green sneezes", etc.

When you hit the node **NOUN**, you are asking the unknown black I called **NOUN** to fetch any noun for you from its storehouse of nouns. This is known as a *procedure call*, in computer science terminology. It means you temporarily give control to a *procedure* (here, **NOUN**) which (1) does thing (produces a noun) and then (2) hands control back to you. In above RTN, there are calls on three such procedures: **ARTICLE**, **ADJECTIVE**, and **NOUN**. Each of these RTN's ORNATE NOUN and the others called from

*Original GEB / RTN reference preserved from Volume 1.*

## 729 runes = 27 X 27

Liber Primus pages 0-2 contain exactly 729 runes. Since  $729 = 27 \times 27$ , Volume 1 writes the complete rune sequence into a square matrix, preserving the original order: left to right, 27 runes per row, until all 729 runes are placed.

No runes are added or removed. The proposal is that the geometry of this square is part of the cipher, not merely a convenient layout.

Working model: 729 runes -> 27x27 matrix -> three-rune structures -> Euler totient transformations -> movement -> three-rune keys -> Mobius phase -> Gematria Primus subtraction mod 29 -> plaintext -> next structural clue.

# The 27x27 matrix

|    | 1       | 2       | 3     | 4     | 5       | 6    | 7     | 8     | 9    | 10    | 11   | 12      | 13     | 14    | 15   | 16      | 17      | 18    | 19      | 20    | 21    | 22    | 23    | 24    | 25   | 26   | 27    |
|----|---------|---------|-------|-------|---------|------|-------|-------|------|-------|------|---------|--------|-------|------|---------|---------|-------|---------|-------|-------|-------|-------|-------|------|------|-------|
| 1  | S/Z     | H       | E     | O     | G       | M    | IA/IO | F 1   | S/Z  | Y     | E    | NG      | C/K 13 | TH    | J    | G       | AE 1    | F     | J       | OE 1  | O     | N     | TH 1  | N     | E    | W    | B     |
| 2  | A       | S/Z     | OE    | EO    | I       | NG   | U/V 1 | N     | AE   | G     | I    | TH 1    | H      | B     | N    | IA/IO 1 | P       | B     | N 1     | EO    | J     | TH 1  | A     | NG    | Y    | X 1  | B     |
| 3  | D       | P 1     | E     | IA/IO | F       | X    | G     | F 1   | S/Z  | P     | T    | F       | P      | U/V   | NG   | Y       | NG      | X     | IA/IO 1 | Y     | H 1   | B     | U/V 1 | G     | O    | P 1  | NG    |
| 4  | AE      | TH 1    | I     | AE    | H       | X    | TH    | P     | G 1  | Y     | F    | NG      | C/K    | J 1   | TH   | A       | M       | I     | NG      | I     | IA/IO | Y 1   | H     | EO 1  | T    | H    | AE    |
| 5  | IA/IO 1 | X       | Y 1   | E     | U/V     | NG 1 | C/K   | F     | N 1  | EO    | TH   | J       | I      | TH 1  | TH   | P 1     | Y       | NG    | EA 1    | Y     | N     | E     | Y 1   | D     | X    | NG   | W 1   |
| 6  | B       | C/K     | X     | D     | B       | F 1  | M     | T     | N    | E 1   | EA   | J       | N      | L     | G    | B 1     | X       | G     | TH 4    | Y     | I     | D     | A 1   | NG    | G    | M    | J     |
| 7  | R       | O       | L     | EO 1  | Y       | T    | P     | I     | TH   | D     | J 1  | OE      | H      | L 1   | EA 1 | F       | X       | AE    | P       | G     | X 1   | F     | L     | W     | EO   | T    | AE 1  |
| 8  | F       | G       | N 1   | M     | EO      | L    | N 1   | NG    | M    | F     | R    | IA/IO 1 | A      | S/Z 1 | EA   | M       | NG      | X     | X       | EO    | D     | B 1   | OE    | D     | M    | O 1  | EA    |
| 9  | EO      | H 1     | D     | NG    | G 1     | OE   | NG    | L     | U/V  | R     | N    | T 1     | AE     | S/Z   | Y    | U/V     | H       | R     | T 1     | H     | C/K 1 | S/Z   | OE    | T     | OE   | NG   | U/V   |
| 10 | R 1     | S/Z 1   | F     | O     | E       | W    | EA    | OE    | L    | F     | AE 1 | M       | R      | NG 1  | D    | A       | M       | R 1   | W       | A     | I     | M     | S/Z   | N 1   | S/Z  | OE   | R     |
| 11 | U/V     | W       | S/Z   | L     | IA/IO 4 | OE   | A     | AE    | NG   | S/Z   | D    | P       | T 1    | C/K   | R    | TH      | IA/IO 1 | R     | B       | O     | D     | TH    | F 1   | Y     | X    | I    | T 4   |
| 12 | OE      | I 1     | F     | L     | O 1     | F    | EA 1  | R     | O    | OE    | M    | H       | M      | G     | P    | H 1     | AE      | H     | N       | O     | H     | Y 1   | OE    | S/Z   | L 1  | N    | G 1   |
| 13 | AE      | Y 1     | OE    | C/K   | B       | L    | J 1   | NG    | L    | OE 1  | AE   | J       | EA     | W 1   | EA   | TH      | O 1     | B     | OE      | Y 1   | C/K   | F     | C/K   | J 1   | L    | AE 4 | N 1   |
| 14 | TH      | P 1     | U/V   | X 1   | OE      | X    | G 1   | P     | F    | S/Z   | EO   | AE      | OE 1   | NG    | P    | EO      | O       | E     | A 1     | G     | AE    | IA/IO | NG    | TH    | O 1  | P    | A     |
| 15 | OE      | TH      | R     | NG    | AE 1    | C/K  | S/Z   | B     | EO   | Y     | H 1  | T       | X      | E     | L    | R 4     | G       | W     | Y 1     | J     | F     | I     | N     | IA/IO | C/K  | Y    | EA    |
| 16 | I       | IA/IO 1 | O     | TH 1  | E       | C/K  | AE    | C/K   | X    | IA/IO | EA 1 | O       | L      | C/K 1 | F    | R       | D       | NG    | E       | U/V 1 | D     | C/K   | L     | EA    | S/Z  | X    | C/K   |
| 17 | G       | IA/IO 4 | W     | S/Z   | TH 1    | F    | D     | NG 1  | I    | IA/IO | M    | A       | AE     | G 1   | W    | S/Z 1   | N       | D 1   | C/K     | P     | TH    | X     | P     | EA    | EA 1 | W    | U/V 1 |
| 18 | EA      | W 1     | F     | W     | J       | Y 1  | X     | D     | W    | C/K   | G    | C/K     | OE 1   | D     | X    | OE 1    | R       | IA/IO | G 1     | N     | P     | A     | Y     | P 1   | C/K  | Y    | H 1   |
| 19 | F       | E       | J     | EA    | N 1     | OE   | AE 1  | U/V   | A 1  | H     | R 1  | E       | EA     | TH    | F    | J       | A 1     | L     | X       | S/Z   | T 1   | M     | F     | L     | F    | T 1  | G     |
| 20 | I       | TH 1    | U/V   | L     | G 1     | X    | EA    | T     | S/Z  | L     | J    | P 1     | L      | X     | I    | OE      | M 1     | U/V 4 | H       | T 1   | B     | EO    | L     | D     | H    | B    | M 1   |
| 21 | EA      | R       | B 1   | N     | H       | B    | E     | G     | EO 1 | D     | L    | W       | EO     | IA/IO | P    | O 1     | H       | E     | EA 1    | W     | I     | R     | I     | H 1   | U/V  | TH   | H     |
| 22 | Y 1     | N       | X     | B     | G       | J    | P     | U/V 1 | NG   | EA    | F    | N       | I      | E     | D    | IA/IO   | NG      | R 1   | D       | B     | J     | IA/IO | OE    | M     | I 1  | F    | T 1   |
| 23 | J       | D       | I     | TH    | R       | L    | S/Z 1 | E     | EO   | O     | G    | B       | T      | D 1   | TH   | A       | N       | C/K   | Y 1     | IA/IO | S/Z   | TH    | D 1   | NG    | F    | L    | E     |
| 24 | G       | H       | C/K 1 | E     | O       | I    | T     | N     | X 1  | P     | T    | F       | H      | R     | D    | E       | F       | T 1   | AE      | W     | H 1   | B     | C/K 1 | F 1   | P    | A    | L     |
| 25 | U/V     | EA      | N     | L     | J 1     | J    | C/K   | L     | W    | EA    | D    | U/V     | D      | EO 1  | EA   | X       | D       | W     | H       | EA 1  | TH    | IA/IO | AE    | C/K   | L    | T    | W     |
| 26 | E       | I       | C/K 1 | P     | OE      | D    | C/K 1 | N     | H    | A 1   | R    | I       | G      | TH    | EA   | E       | T       | G 1   | TH      | H     | O     | IA/IO | W     | AE    | J    | E 1  | NG    |
| 27 | EA      | D 1     | O     | AE 1  | A       | L 1  | B     | J     | C/K  | U/V   | X    | T       | A      | B     | J    | P 1     | F       | EA 1  | H       | D     | N     | IA/IO | U/V   | P     | S/Z  | U/V  | W 13  |

Original matrix image retained. The highlighted geometry is part of the working research map.

## STAGE 1

# AE-J-EA -> AE-I-EA

The first important three-rune structure is AE-J-EA. Its center is J = 11.

$\text{phi}(11) = 10$  Gematria Primus value 10 = I AE-J-EA -> AE-I-EA

So the generated repeating key is AE-I-EA. The same transformed value is also reused to derive movement:

J ->  $\text{phi}(J) = 10 = I$   $\text{phi}(10) = 4$  10 + 4 = 14 RIGHT 14

This is the first clear example of the central idea in Volume 1: the value obtained from a node is not used only once. The same totient chain produces both the key and the movement that leads to the ciphertext.

# AS I GO THE

At the resulting position, the encrypted runes are:

L - AE - N - TH - P - U - X

Repeat the three-rune key across seven positions:

AE - I - EA - AE - I - EA - AE

Using the Volume 1 rule  $P = C - K \bmod 29$ :

|            |    |    |    |    |   |    |    |
|------------|----|----|----|----|---|----|----|
| ciphertext | L  | AE | N  | TH | P | U  | X  |
| key        | AE | I  | EA | AE | I | EA | AE |
| result     | A  | S  | I  | G  | O | TH | E  |

# AS I GO THE

The segment ends on the ciphertext rune X, which immediately becomes part of the next structure.

## STAGE 2

# X-OE-X -> X-I-X -> WEATHER

The final X opens an adjacent mirrored structure:

X-OE-X

Its center is OE. Volume 1 applies the same center transformation:

$\text{phi}(OE) = 10 = I \quad X-OE-X \rightarrow X-I-X$

The same value 10 is then used as the next movement: RIGHT 10. This lands on NG at the exact center of the 27x27 matrix.

The new key is X-I-X, but its starting phase must still be determined. That is where the Mobius rule enters.

## WEATHER

In the original PDF the five ciphertext runes are shown visually in the matrix rather than printed as a separate line. Volume 1 gives the active five-position key pattern X-X-I-X-X and the proposed plaintext WEATHER.

# Mobius selects the rotation

A repeating three-rune key can begin in three cyclic phases. Volume 1 proposes a deterministic rule so the phase is not chosen by whichever rotation looks most like English:

$$p = \sum \mu(\phi(K_i)) \bmod 3$$

$p = 0 \rightarrow$  start from rune 1

$p = 1 \rightarrow$  start from rune 2

$p = 2 \rightarrow$  start from rune 3

| key     | $\phi(K)$ | $\mu$     | phase | active start |
|---------|-----------|-----------|-------|--------------|
| AE-I-EA | 20, 4, 12 | 0, 0, 0   | 0     | AE-I-EA      |
| X-I-X   | 6, 4, 6   | +1, 0, +1 | 2     | X-X-I        |
| H-NG-C  | 4, 12, 4  | 0, 0, 0   | 0     | H-NG-C       |
| H-U-H   | 4, 1, 4   | 0, +1, 0  | 1     | U-H-H        |

For X-I-X, the signature 6-4-6 gives phase 2, so the active key begins X-X-I. This is the phase used for WEATHER.



## A(14,19): the crossroads

After WEATHER, Volume 1 continues from A(14,19). The two already derived control values are I = 10 and  $\phi(I) = 4$ . They are reused as movement values.

```
UP 10 -> NG RIGHT 4 -> NG(14,23)
```

The right-hand NG is placed symmetrically between A cells four positions away:

```
A(14,19) --4-- NG(14,23) --4-- A(14,27)
```

From NG(14,23), the established value 10 is used vertically:

```
UP 10 -> H(4,23) DOWN 10 -> C(24,23)
```

Together these positions form the non-mirrored key H-NG-C. Its totient signature is 4-12-4, the same signature Volume 1 notes for the mirrored structure I-NG-I.

# TURNS

Reading upward from A gives:

A-OE-N-B-W

Use the repeating key:

H-NG-C-H-NG

Mod-29 subtraction produces:

A-OE-N-B-W - H-NG-C-H-NG = T-U-R-N-S

# TURNS

At this point the center of H-NG-C becomes important:  $NG = 21$ , so  $\phi(NG) = 12$ . Volume 1 uses this single value in two directions from the same A crossroads.

## $\phi(\text{NG}) = 12$ : two branches

From  $A(14,19)$ , the value 12 is used twice:

DOWN 12  $\rightarrow$  center of H-TH-H LEFT 12  $\rightarrow$   $G(14,7)$ , the start of the next ciphertext

This is one of the cleanest transitions in the Volume 1 route: one previously derived number points to both the next key-generating structure and the next ciphertext start.

The downward branch reaches the mirrored node H-TH-H. Apply the standard center transformation:

$\text{TH} = 2 \quad \phi(2) = 1 = \text{U} \quad \text{H-TH-H} \rightarrow \text{H-U-H}$

For H-U-H, the totient signature is 4-1-4. The Mobius values are 0, +1, 0, giving phase 1. Therefore the active key begins U-H-H.

# COLD

The left branch lands at G(14,7). Reading upward gives the four-rune ciphertext:

G-J-EA-A

Using the phase-1 repeating key:

U-H-H-U

The subtraction is:

G-J-EA-A - U-H-H-U = C-O-L-D

# COLD

The four-stage proposed plaintext is therefore:

**AS I GO, THE WEATHER TURNS COLD.**

# Totient signatures returning

Volume 1 repeatedly compares structures through their totient signatures. The important point is not that every node is identical, but that different rune structures can reduce to the same numerical family.

|          |   |         |
|----------|---|---------|
| X-I-X    | → | 6-4-6   |
| H-NG-C/K | → | 4-12-4  |
| EA-H-EA  | → | 12-4-12 |
| H-U-H    | → | 4-1-4   |

*Original working note: rune structures and their totient signatures.*

```

6 - 4 - 6
  ↓
4 - 12 - 4
  ↓
12 - 4 - 12
  ↓
4 - 1 - 4

```

*Original working note: one observed numerical inheritance chain.*

These relationships are supporting structure, not a replacement for the direct decryption arithmetic. They are useful because the same values reappear as keys, phases, and movement distances.

## 233, 51, columns 19 and 7

Volume 1 records several numerical observations around the core route. They are presented as supporting evidence rather than required steps.

21 plaintext runes. AS I GO THE = 7 runes, WEATHER = 5, TURNS = 5, COLD = 4. Total: 21.

Sum = 233. The 21 zero-based Gematria Primus indices of the proposed plaintext sum to 233, which Volume 1 notes as a Fibonacci number.

COLD = 51. The zero-based indices C(5)+O(3)+L(20)+D(23) sum to 51.

Columns 19 and 7. Volume 1 notes W has zero-based GP index 7 and prime value 19, so  $19 - 7 = 12 = \text{phi(NG)}$ . Geometrically, TURNS ends on W in column 19 while COLD begins in column 7.

These numerical relationships are observations around the route. The core decryption remains the four-stage chain derived from nodes, totients, Mobius phase, movement and mod-29 subtraction.

# Transition cells and thematic clues

The source also pays attention to the final ciphertext cell of each stage as a possible hint toward the next step:

After AS I GO THE: the final X opens the adjacent X-OE-X mirror.

After WEATHER: the visual route reaches A(14,19), the crossroads.

After TURNS: the final W lies in column 19 and is linked to the proposed 19 -> 7 transition through  $\phi(\text{NG})=12$ .

After COLD: the endpoint A(11,7) gives  $\phi(A)=8=H$ , noted as an additional hint toward H.

Volume 1 also records the separator/dot geometry, H-TH-H uniqueness in the lower region, ROAD/PATH/WAY, prime 19 observations, DIVINITY WITHIN = 491 / "A" CROSSROADS = 491, and other thematic relationships. These are exploratory and are not needed to reproduce the four plaintext stages.

# The route in one sequence

AE-J-EA -> AE-I-EA -> RIGHT 14 -> AS I GO THE -> X-0E-X -> X-I-X -> Mobius phase 2  
= X-X-I -> RIGHT 10 -> central NG -> WEATHER -> A(14,19) crossroads -> reuse 10 and  
4 -> H-NG-C -> TURNS ->  $\phi(NG)=12$  -> DOWN 12 / LEFT 12 -> H-TH-H -> H-U-H ->  
Mobius phase 1 = U-H-H -> COLD

## AS I GO, THE WEATHER TURNS COLD.

This is the strongest result recorded in Volume 1. It should be treated as a proposed cryptanalytic construction: reproducible from the stated operations, but not an officially verified Cicada 3301 solution.



# What Volume 1 establishes

The central claim of Volume 1 is not simply that readable English can be extracted. It is that the same small set of ideas repeatedly reconnects the route: three-rune structures, Euler totients, inherited movement values, Mobius phase selection, and Gematria Primus subtraction.

The most important next test is whether the same structural framework can continue beyond COLD without introducing arbitrary rules.

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Reader edition · rebuilt from the original Volume 1 text and preserved working images.